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Project Title

Listening to Farmers: NLP and Large Language Model Evidence on the Awareness–Utilisation Gap in Welfare Schemes from Chittoor District, Andhra Pradesh

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Abstract

Government welfare schemes for Indian farmers are evaluated almost entirely through structured satisfaction ratings, although farmers also describe their experiences in open-ended answers that are seldom analysed. This study analyses a field survey of 1,500 farmers from Chittoor district, Andhra Pradesh, verified by the Ethnic Community, Government of Andhra Pradesh, and introduces an **Awareness–Utilisation–Delivery–Experience (AUDE) framework** that measures where scheme benefits are lost between knowing about a scheme and benefiting from it. Structured answers measure awareness and enrolment, while Natural Language Processing (NLP) of 1,214 written responses measures delivery and experience through aspect-based sentiment analysis, topic modelling, suggestion mining and rating–text triangulation.

Although 73.7% of farmers know at least one scheme, awareness converts unevenly into enrolment, from 89.4% for Rythu Bharosa to 9.1% for Crop Insurance and 2.2% for PM Krishi Sinchai Yojana. Payment delays account for about 44% of complaint text, and enrolment ease and timeliness are among the most negatively described service aspects. Most strikingly, 87% of farmers rate their scheme positively, yet only 24% describe it positively in their own words when that text is classified by the best validated model; 771 of 1,031 positive raters write mixed, neutral or negative accounts (McNemar–Bowker $\chi^2 = 705$, $p < 0.001$). Satisfaction ratings therefore substantially overstate how farmers experience scheme delivery.

To establish which methods can be trusted on such text, eight representative sentiment approaches were benchmarked against 243 human-labelled responses. A fine-tuned ModernBERT encoder achieved the highest macro-F1 (0.765), statistically tied with a frontier large language model (0.742), while the widely used VADER lexicon reached only 0.223. An Indian agricultural LLM, Dhenu2, outperformed a general LLM of identical size by 0.313 macro-F1. All corpus-level findings are reported with this validated model, so they do not depend on the unreliable lexicon. The findings support evaluating schemes through farmers' own words, but only with models validated on in-domain labels.

Keywords: natural language processing; aspect-based sentiment analysis; large language models; agricultural policy; welfare schemes; awareness–utilisation gap; survey triangulation; India

1. Introduction

Agriculture employs nearly half of India's workforce, and central and state governments operate numerous welfare schemes, including PM-KISAN, Rythu Bharosa, crop insurance, minimum support price mechanisms and input subsidies, to stabilise farm incomes. The value of these schemes depends on a chain of conditions: farmers must know about a scheme, be able to enrol, receive benefits on time, and experience the support as useful. A failure at any stage reduces the benefit that reaches the farmer, regardless of the amount of public money allocated.





In Andhra Pradesh, many schemes are delivered through village-level Rythu Bharosa Kendrams [19], where farmers register, obtain information and receive inputs. Scheme performance is conventionally assessed by asking farmers to rate their satisfaction on a fixed scale. Ratings are convenient to collect, but they compress lived experience into a narrow range and are susceptible to acquiescence and social-desirability bias. A farmer may rate a scheme as satisfactory while, in the same interview, describing delayed payments and repeated document verification.



Open-ended survey answers capture this experience directly, and recent advances in NLP make it feasible to analyse them at scale. Text-as-data methods have become established in political science and policy research [5, 6], and large language models (LLMs) now approach human accuracy when coding open-ended survey responses [12, 13]. Their reliability on rural, translated and code-switched farmer feedback, however, has not been established. This study combines structured survey measures with validated NLP to trace where welfare-scheme benefits are lost, and evaluates which NLP methods can be trusted for that purpose.

2. Literature Review

2.1 Awareness and utilisation of farmer welfare schemes

Studies of Indian farmer welfare schemes rely predominantly on structured surveys of awareness and perception. A recent survey of PM-KISAN beneficiaries in Bhagalpur district associated greater scheme awareness with more stable income growth and identified digital literacy and procedural complexity as persistent barriers to full utilisation [9]. Such studies report percentages of awareness and satisfaction but rarely examine the reasons farmers give in their own words, and none identified measures where along the path from awareness to benefit the losses occur.

2.2 Sentiment analysis and text mining in agriculture and policy

Sentiment analysis and opinion mining are mature fields [1, 2], yet text mining remains comparatively underused in agricultural research [7]. In the Indian agricultural context, sentiment classification has been applied mainly to social-media discussion, such as the farmers' protest on Twitter, using TF-IDF features with classical machine learning [8]. Lexicon methods such as VADER [3] and topic models such as Latent Dirichlet Allocation [4] are widely used in policy research [6], but their output is often reported without validation against human judgement, a practice cautioned against in the text-as-data literature [5].

2.3 NLP for open-ended survey responses

Aspect-based sentiment analysis has been applied successfully to open-ended employee-satisfaction surveys, where few-shot transformer models outperformed bag-of-words and zero-shot baselines [10]. In healthcare, the validated AI-PREM tool reported high F1 for positive patient feedback but only 0.63 for negative feedback, showing that methods can perform unevenly across sentiment classes [11]. LLMs have outperformed crowd workers on several annotation tasks [12], and on open-ended survey responses the best LLM approached the accuracy of a second human coder [13]. Modern encoders such as ModernBERT [15] and DeBERTa-v3 [23], few-label methods such as SetFit [16], general LLMs such as Qwen2.5 [17] and domain LLMs such as the Indian agricultural model Dhenu2 [18] offer several alternatives, but they have not been compared on farmer feedback.

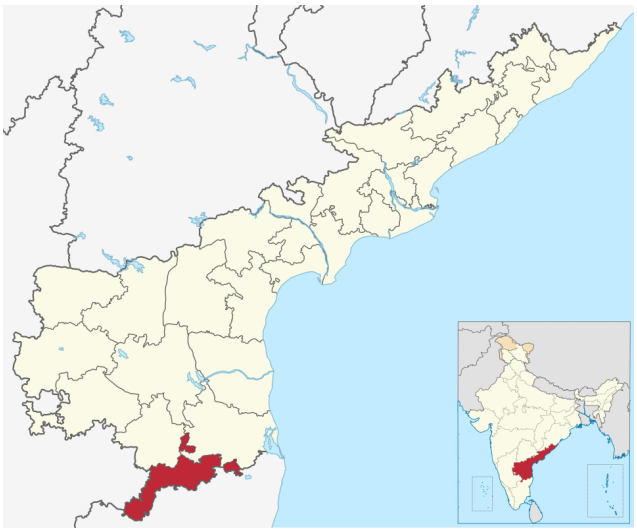
2.4 Research gap

- Scheme evaluations measure awareness or satisfaction separately; no framework traces benefit loss across awareness, enrolment, delivery and experience within the same respondents.
- No study compares satisfaction ratings with validated NLP sentiment of the same farmers' written feedback to measure how far ratings overstate their experience.
- Sentiment tools are applied to farmer text without validation, and no benchmark compares lexicon, transformer, few-label, fine-tuned and LLM approaches on farmer feedback.
- Whether agricultural domain knowledge improves LLM interpretation of farmer feedback has not been tested against a general model of the same size.

2.5 Contributions

1. The **AUDE framework**, which quantifies benefit loss at four stages using structured answers and farmers' own words from the same interviews.
2. The **satisfaction gap**: a validated, statistically tested measure of how far satisfaction ratings overstate farmers' written experience, overall and by scheme.
3. A **human-validated benchmark** of eight sentiment approaches on Indian farmer feedback, including the first controlled comparison of an agricultural LLM with a general LLM of identical size.
4. A **policy priority matrix** and evidence-linked recommendations for the Department of Agriculture and Farmers Welfare.

3. Study Area and Data



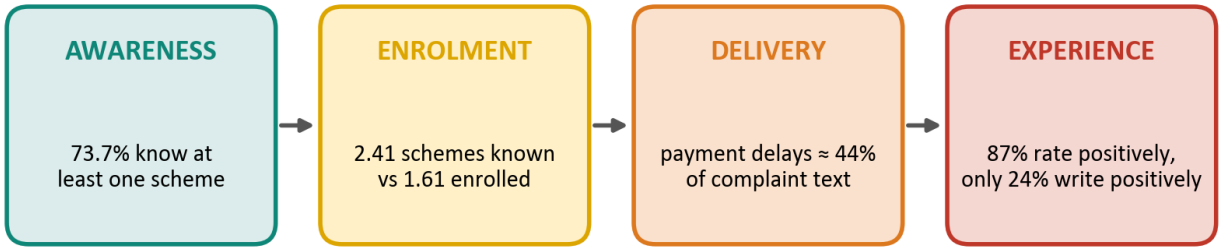
Chittoor district lies in the Rayalaseema region of southern Andhra Pradesh, bordering Tamil Nadu and Karnataka. Agriculture in the district is dominated by smallholders cultivating mango, paddy and sugarcane under largely rain-fed conditions.

Data were collected from 1,500 farmers within a 50 km radius of the university campus through structured interviews conducted by trained investigators, and the dataset was verified by the Ethnic Community, Government of Andhra Pradesh. Interviews were conducted mainly in Telugu (90.5%), and answers were recorded in English.

The instrument covered five sections: farmer profile (12 questions), awareness of schemes (5), scheme feedback including ratings of six service aspects and three open-ended answers on positive aspects, problems and suggestions (11), outlook (4) and interview details (4). The mean age of respondents was 50.7 years and the mean farming experience 19.8 years; 67.2% were male and 32.8% female. The median landholding was 2.05 acres, and 53.3% of farmers were marginal (below 2.5 acres). In total, 1,214 farmers provided written feedback on a scheme.

3.1 Conceptual framework

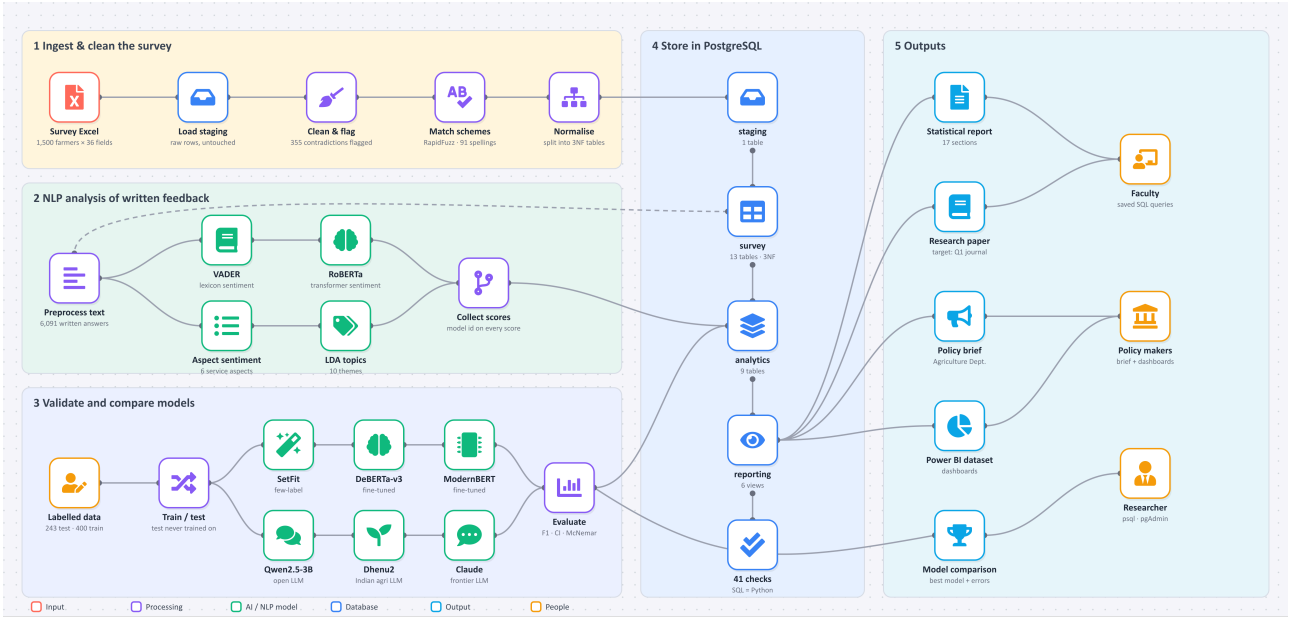
The study is organised around four stages at which scheme benefits can be lost. Awareness and enrolment are measured from structured answers; delivery and experience are measured from farmers' written feedback using NLP. Each stage is quantified within the same set of respondents.



Measured from structured answers (awareness, enrolment) and from farmers' own words via NLP (delivery, experience)

4. Methodology

The analysis follows a reproducible pipeline that converts the raw survey into validated findings, from ingestion and cleaning through NLP analysis and model validation to the generation of outputs.



4.1 Text preprocessing

The three open-ended answers of each farmer were cleaned and normalised. Answers consisting only of non-informative phrases such as "No", "Nothing" or "No problems" were treated as empty rather than as sentiment, since lexicon methods otherwise score them as negative. Scheme names, which appeared in 93 different spellings, were mapped to 11 canonical schemes by combining an abbreviation dictionary with fuzzy string matching.

src/preprocess.py — lines 47–62

```python
def normalize_scheme_name(raw_name, choices=CANONICAL_SCHEMES, score_cutoff=70):
    text = clean_text(raw_name)
    if text is None or is_nil_response(text):
        return None

    first_segment = re.split(r"[,/&]| and ", text)[0].strip() or text

    abbreviation = SCHEME_ABBREVIATIONS.get(first_segment.lower())
    if abbreviation:
        return abbreviation

    match = process.extractOne(
        first_segment, choices, scorer=fuzz.WRatio,
        processor=default_process, score_cutoff=score_cutoff,
    )
    return "Other / Unmatched" if match is None else match[0]
```

4.2 Analysis of written feedback

- **Aspect-based sentiment analysis** splits each answer into clauses, detects which of the six rated service aspects each clause refers to, and scores the sentiment of that clause, so that a single answer can be positive about the amount and negative about timeliness.
- **Topic modelling** with Latent Dirichlet Allocation [4] identifies recurring themes separately in complaints and in positive feedback.

- **Suggestion mining** ranks the changes farmers request most frequently across 927 substantive suggestions.
- **Rating–text triangulation** compares each farmer's satisfaction rating with the sentiment of the same farmer's written feedback, and the McNemar–Bowker test [21] examines whether the two are systematically asymmetric.
- **Subgroup analysis** uses chi-square tests with Cramér's V to relate sentiment to education, income, landholding, gender, information source and scheme.

src/absa.py — lines 44–72

```python
def extract_aspects(text):
    if pd.isna(text) or not text:
        return set()
    lowered = text.lower()
    return {aspect for aspect, pattern in _ASPECT_PATTERNS.items() if pattern.search(lowered)}


def score_aspects(text, analyzer, score_fn):
    if pd.isna(text) or not text:
        return {}

    clauses = [c.strip() for c in _CLAUSE_SPLIT.split(text) if c.strip()] or [text]

    scores_by_aspect = {}
    for clause in clauses:
        aspects = extract_aspects(clause)
        if not aspects:
            continue
        score, _ = score_fn(analyzer, clause)
        if score is None:
            continue
        for aspect in aspects:
            scores_by_aspect.setdefault(aspect, []).append(score)

    return {aspect: sum(s) / len(s) for aspect, s in scores_by_aspect.items()}


aspect_label = label_from_score
```

src/triangulate.py — lines 33–46

```python
def flag_divergence(nlp_label, rating_polarity):
    if pd.isna(nlp_label) or pd.isna(rating_polarity):
        return None
    return {nlp_label, rating_polarity} == {"Positive", "Negative"}


def add_overall_divergence(df, overall_label_col):
    df = df.copy()
    df["divergence_overall"] = [
        flag_divergence(nlp, rating)
        for nlp, rating in zip(df[overall_label_col], df["C2_satisfaction_polarity"])
    ]
    return df
```

4.3 Model selection

Eight sentiment approaches were selected to represent the main families used in applied research, so that each comparison answers a specific question rather than adding models for their own sake:

- **VADER** [3], the most common lexicon tool in applied studies, extended with an agricultural word list.
- **RoBERTa** trained on social-media text [22], representing pre-trained transformers applied without adaptation.
- **TF-IDF with logistic regression**, the classical supervised approach used in earlier agricultural sentiment work [8].
- **SetFit** [16], a few-label method designed for settings with only hundreds of labelled examples.

- **ModernBERT** [15], a recent encoder fine-tuned on the labelled farmer responses.
- **Dhenu2** [18] and **Qwen2.5-3B** [17], an Indian agricultural LLM and a general LLM with the same 3-billion-parameter size, forming a controlled test of domain knowledge.
- **Claude**, a frontier LLM, as an upper reference for prompting without training.

The LLMs received the same written labelling guidelines given to the human annotators, either alone or with three labelled examples per class, and were decoded deterministically.

src/model_comparison.py — lines 17–24

```
17 LLM_INSTRUCTIONS = (
18     "You label farmers' feedback about Indian government welfare schemes. "
19     "Reply with exactly one word: Positive, Negative or Neutral.\n"
20     "Positive = mainly expresses satisfaction, benefit, relief or approval.\n"
21     "Negative = mainly expresses a problem, delay, frustration or complaint.\n"
22     "Neutral = factual with no clear tone, or genuinely mixed with no dominant tone."
23 )
```

4.4 Validation protocol

A stratified random sample of 243 responses (20% of those with feedback) was labelled as Positive, Neutral or Negative and held out as the test set; it was never used for training or for selecting examples. A separate training set of 400 responses was drawn half at random and half from responses on which existing models were least certain. Labels were produced jointly by the author and one additional annotator using a written guide; for the training set, labels were first suggested by an AI model and then verified, and 86 of the 400 suggestions (21.5%) were changed. Trainable models were run with five random seeds. Performance is reported as macro-F1, which weights the three classes equally, with 95% bootstrap confidence intervals, and differences between models are tested with the exact McNemar test [20].

A1 | RespondentID

	A	B	D	E	F	G
1	RespondentID	Source_Excel_Row	Scheme	Feedback_Text	Human_Label	Annotator_Initials
2	R0001	365	Kisan Credit Ca	Money		
3	R0002	1271	Input Subsidy	Reduce financial burden. some schemes are not applicable for farmers who are doing farming on leased land. timely release		
4	R0003	1342	Rythu Bharosa	Financial support helped reduce farming expenses. Sometimes the amount is not credited. provide multiple stores to easy accesss of the farming resources		
5	R0004	1364	Rythu Bharosa	Government assistance reduced financial pressure during drought conditions. Delay in payments. Ensure all deserving farmers receive benefits		
6	R0005	1361	Unspecified	Timeliness. Delay of benifits. Information should be delivered and provide clear eligibility information		
	R0006	152	Rythu Bharosa	Dealt in payments. Ensuring market support		

4.5 Corpus-wide scoring with validated models

After the benchmark, all 1,214 written responses were rescored with the best model so that every reported finding rests on a validated classifier rather than on the first-stage lexicon. ModernBERT was retrained with five seeds on the training set and the majority label was used; on the held-out test set this ensemble reached a macro-F1 of 0.760 and an accuracy of 83.5%. For aspect-level sentiment, two pipelines were compared on 274 human-labelled aspect mentions: keyword detection with a dedicated DeBERTa aspect model (macro-F1 0.379, $\kappa = 0.264$) and keyword detection with ModernBERT applied

to the relevant clauses (macro-F1 0.367). The DeBERTa pipeline was retained. Because aspect-level accuracy remains modest, aspect findings are reported only where they agree with the independent topic-model evidence.

5. Implementation

The system is implemented in Python 3.12. Data handling uses pandas, openpyxl and RapidFuzz; lexicon sentiment uses NLTK VADER; topic modelling and classical models use scikit-learn; transformer models use Hugging Face Transformers and SetFit; and the 3-billion-parameter LLMs run on an NVIDIA RTX 2050 laptop GPU with 4-bit quantisation using PyTorch and bitsandbytes. Statistical tests use SciPy. Survey responses, model outputs and human labels are stored in PostgreSQL 16, and the code is version-controlled on GitHub.

5.1 Project structure

```
FarmerSchemeNLP
├── data/        # raw survey (never modified)
│   └── Research_farmers_dataset.xlsx
├── db/          # PostgreSQL schema + queries
│   ├── queries.sql
│   └── schema.sql
├── docs/        # report, presentation, figures
│   ├── Complete_Project_Report.pdf
│   ├── figures/
│   └── Project_Demonstration.pptx
├── notebooks/   # Colab GPU run
│   └── colab_model_comparison.ipynb
├── output/      # generated results
│   ├── ABSA_Annotation_Worksheet.xlsx
│   ├── Annotation_Worksheet.xlsx
│   ├── claude_test_predictions.csv
│   └── ... 20 more
├── reports/     # statistical report, paper, brief
│   ├── Full_Statistical_Analysis.docx
│   ├── Policy_Intelligence_Brief.md
│   └── ... 3 more
├── results/     # model comparison summary
│   ├── model_comparison.csv
│   └── model_comparison_summary.xlsx
├── scripts/     # pipeline, run in order
│   ├── 01_run_pipeline.py
│   ├── 02_run_analysis.py
│   ├── 03_build_powerbi_dataset.py
│   ├── 04_build_annotation_worksheets.py
│   ├── 05_evaluate_models.py
│   ├── 06_descriptive_statistics.py
│   ├── 07_load_database.py
│   ├── 08_build_statistical_report.py
│   ├── 09_build_research_paper.py
│   ├── 10_build_policy_brief.py
│   ├── 11_build_training_worksheets.py
│   └── 12_compare_models.py
├── src/         # reusable Python modules
│   ├── __init__.py
│   ├── absa.py
│   ├── active_learning.py
│   └── ... 18 more
├── README.md
└── requirements.txt
```

- **data/** holds the original survey workbook, which is never modified.
- **src/** contains 21 reusable modules: text cleaning and scheme matching (preprocess, schemes), sentiment (sentiment_lexicon, sentiment_transformer), aspect analysis (absa), topics (topic_model), suggestions, rating–text triangulation (triangulate), statistical tests, evaluation metrics, selection of training responses (active_learning) and the model benchmark (model_comparison).
- **scripts/** contains twelve numbered steps that rebuild every result from the raw file in order: cleaning and sentiment (01), statistics and topics (02), dashboard data (03), labelling sheets (04, 11), evaluation (05), descriptive statistics (06), storage (07), reports (08–10) and the model benchmark (12).
- **notebooks/** provides a Google Colab notebook for running the LLMs on a cloud GPU.
- **output/**, **results/**, **reports/** and **docs/** hold generated results, the public benchmark summary, the statistical report and paper draft, and the final deliverables.

5.2 Pipeline execution

The evaluation step scores the pipeline's sentiment models against the human labels, both for overall sentiment and for each service aspect:

```
PS> python scripts\05_evaluate_models.py
                    method   n  accuracy  macro_f1  cohen_kappa
          VADER + agri lexicon 243     0.235     0.223       -0.144
RoBERTa transformer (zero-shot) 243     0.506     0.439        0.103
   Logistic regression (TF-IDF) 243     0.761     0.690        0.557
  Feed-forward network (TF-IDF) 243     0.593     0.303        0.057
          aspect   n  accuracy  macro_precision  macro_recall  macro_f1  cohen_kappa
       timeliness  39     0.436            0.431         0.368     0.369        0.151
           amount  35     0.029            0.011         0.333     0.021       -0.049
  enrollment_ease  33     0.424            0.329         0.278     0.287        0.017
official_behavior   3     0.333            0.333         0.111     0.167        0.000
          clarity  18     0.500            0.519         0.342     0.387        0.236
           impact  73     0.712            0.350         0.569     0.308        0.064
     ALL (pooled) 201     0.468            0.505         0.441     0.440        0.191
Saved C:\Users\Krish Singh\Documents\FarmerSchemeNLP\output\validation_results.xlsx
```

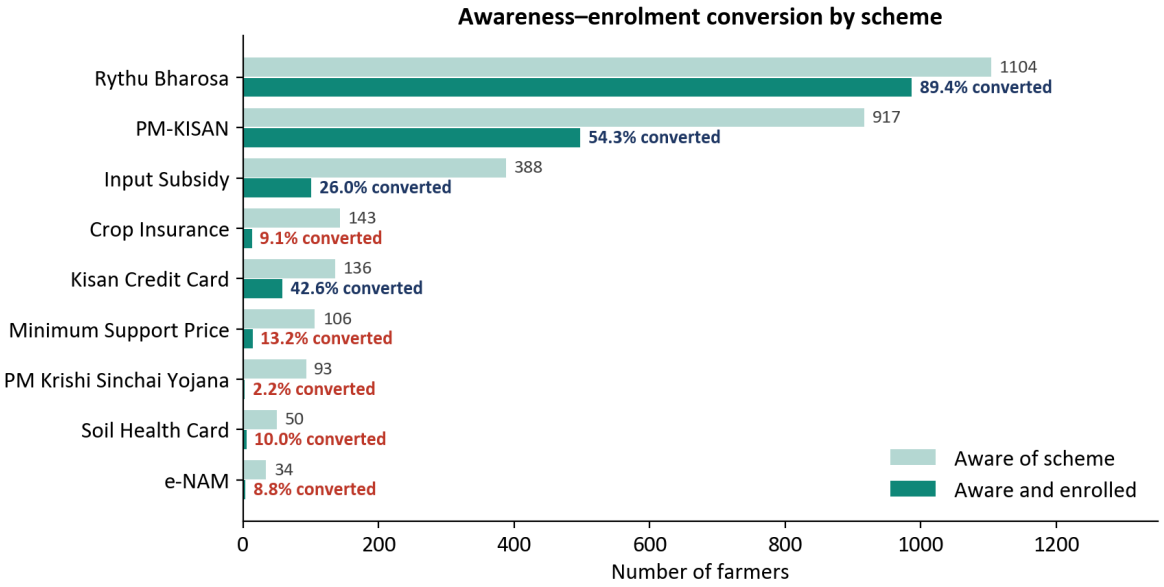
The benchmark step trains or prompts every model, caches its predictions so that interrupted runs can resume, and reports the comparison with confidence intervals:

```
PS> python scripts\12_compare_models.py --llm Qwen/Qwen2.5-3B-Instruct --llm KissanAI/Dhenu2-In
-llm-4bit ...
Test set: 243 labelled rows; training set: 400 labelled rows
  TF-IDF logistic regression: reused from model_predictions.json
  SetFit (all-mpnet-base-v2): reused from model_predictions.json
  Fine-tuned deberta-v3-base: reused from model_predictions.json
  Fine-tuned ModernBERT-base: reused from model_predictions.json
  LLM Qwen2.5-3B-Instruct (zero-shot): reused from model_predictions.json
  LLM Qwen2.5-3B-Instruct (3-shot): reused from model_predictions.json
  LLM Dhenu2-In-Llama3.2-3B-Instruct (zero-shot): reused from model_predictions.json
  LLM Dhenu2-In-Llama3.2-3B-Instruct (3-shot): reused from model_predictions.json
                                      method  accuracy  macro_f1 macro_f1_ci95
                     Fine-tuned ModernBERT-base     0.835     0.765 0.687 - 0.832
                    Claude Opus 5.5 (zero-shot)     0.786     0.742 0.676 - 0.801
   LLM Dhenu2-In-Llama3.2-3B-Instruct (3-shot)     0.749     0.704  0.63 - 0.766
                     SetFit (all-mpnet-base-v2)     0.840     0.675 0.592 - 0.752
                     TF-IDF logistic regression     0.745     0.641 0.562 - 0.718
                     Fine-tuned deberta-v3-base     0.749     0.533 0.476 - 0.595
LLM Dhenu2-In-Llama3.2-3B-Instruct (zero-shot)     0.519     0.445   0.38 - 0.51
                   RoBERTa Twitter (zero-shot)     0.506     0.439 0.367 - 0.512
               LLM Qwen2.5-3B-Instruct (3-shot)     0.457     0.391 0.317 - 0.458
            LLM Qwen2.5-3B-Instruct (zero-shot)     0.300     0.321 0.268 - 0.376
                          VADER + agri lexicon     0.235     0.223 0.163 - 0.279
                                    method  detected  aspect_mentions  macro_f1
 Keyword ABSA + VADER (current pipeline)       201              274     0.373
DeBERTa ABSA (deberta-v3-base-absa-v1.1)       274              274     0.433
```

6. Results

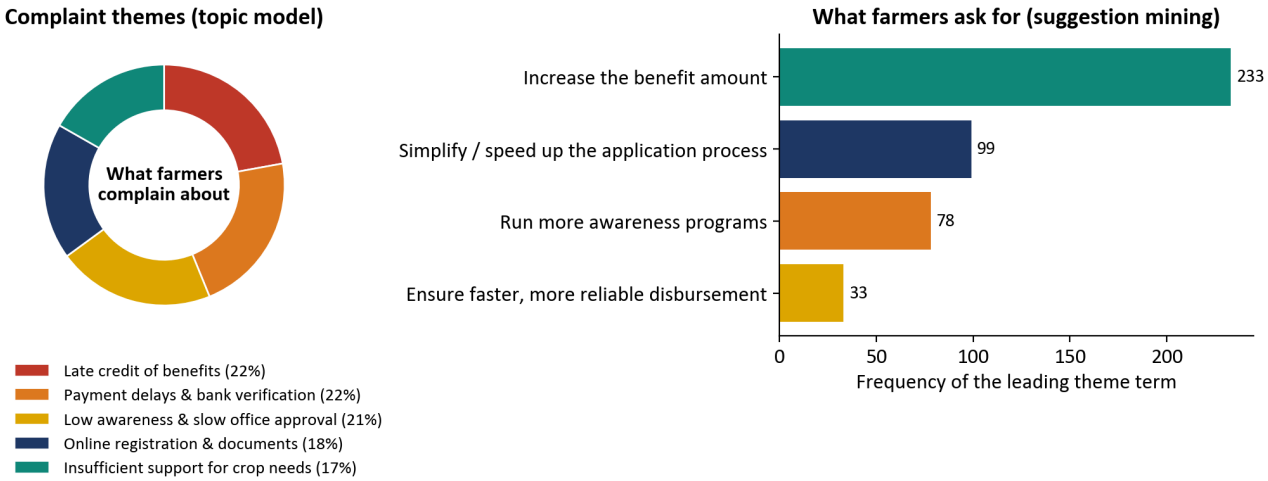
6.1 Awareness and enrolment

73.7% of farmers (1,082 of 1,469) were aware of at least one welfare scheme, leaving 387 farmers (26.3%) unaware of any. Awareness, however, does not translate evenly into utilisation. On average a farmer knew 2.41 schemes but was enrolled in only 1.61, and 51.5% of farmers knew of more schemes than they used.



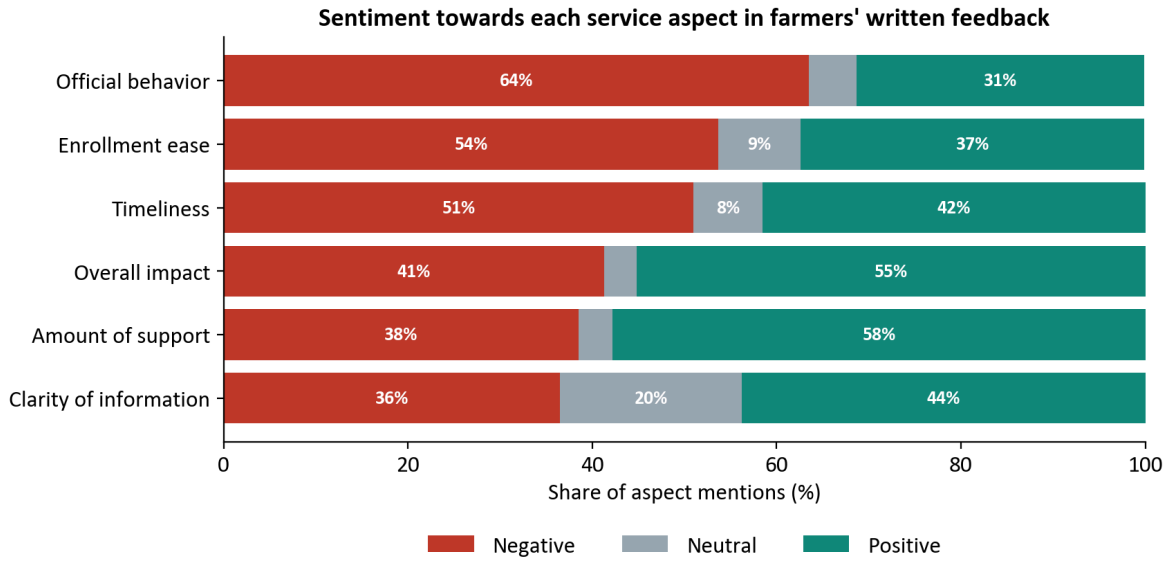
Conversion from awareness to enrolment ranges from 89.4% for Rythu Bharosa, which is delivered through village centres, to 9.1% for Crop Insurance and 2.2% for PM Krishi Sinchai Yojana. For these schemes the binding constraint is enrolment rather than awareness, so further awareness campaigns alone are unlikely to raise utilisation.

6.2 Delivery: what farmers complain about



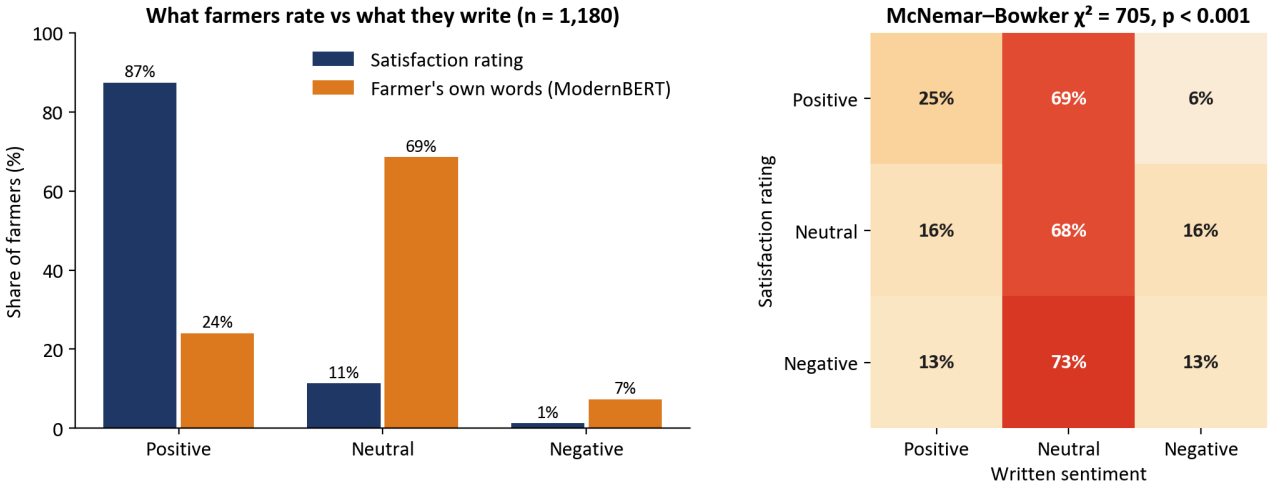
Payment timing dominates complaints: late credit of benefits and payment delays with bank verification together account for about 44% of complaint text, followed by low awareness with slow

office approval and online registration with documentation. Farmers' suggestions, by contrast, place a larger benefit amount first, followed by simpler applications, more awareness programmes and faster disbursement. Delays are the problem farmers experience; a larger amount is the change they most request.



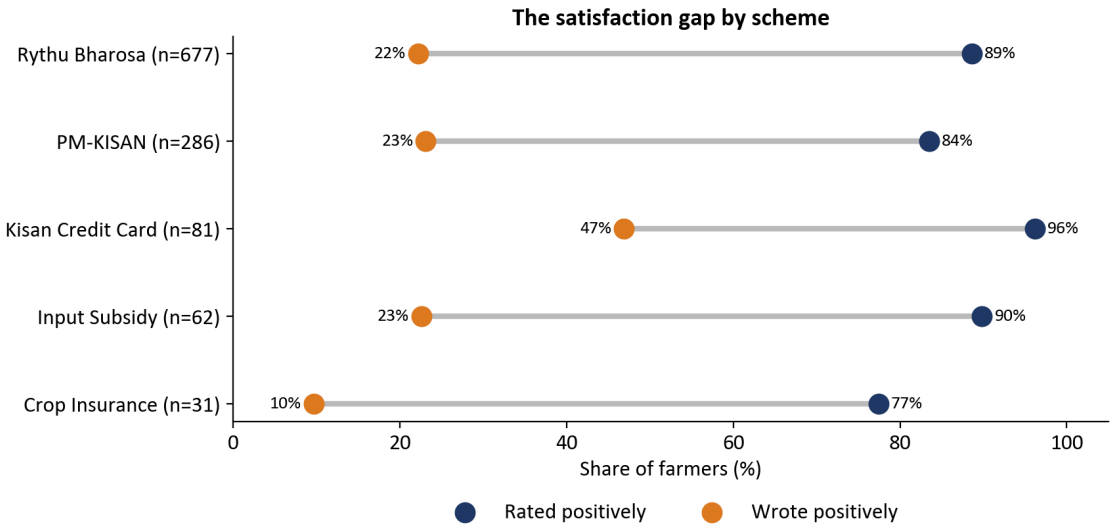
Across 2,659 aspect mentions, enrolment ease (53.7% negative) and timeliness (51.0% negative) are the most criticised of the frequently mentioned aspects, while the amount of support (57.8% positive) and overall impact (55.2% positive) are described mostly favourably. The same two aspects were also the most negative under the first-stage keyword method (45.1% and 39.9%), and they match the dominant complaint topics, so this ranking is robust to the choice of method. Official behaviour is described negatively in 63.5% of its 96 mentions, but this aspect is rarely raised and differed sharply between methods, so it is treated as indicative only.

6.3 Experience: the satisfaction gap



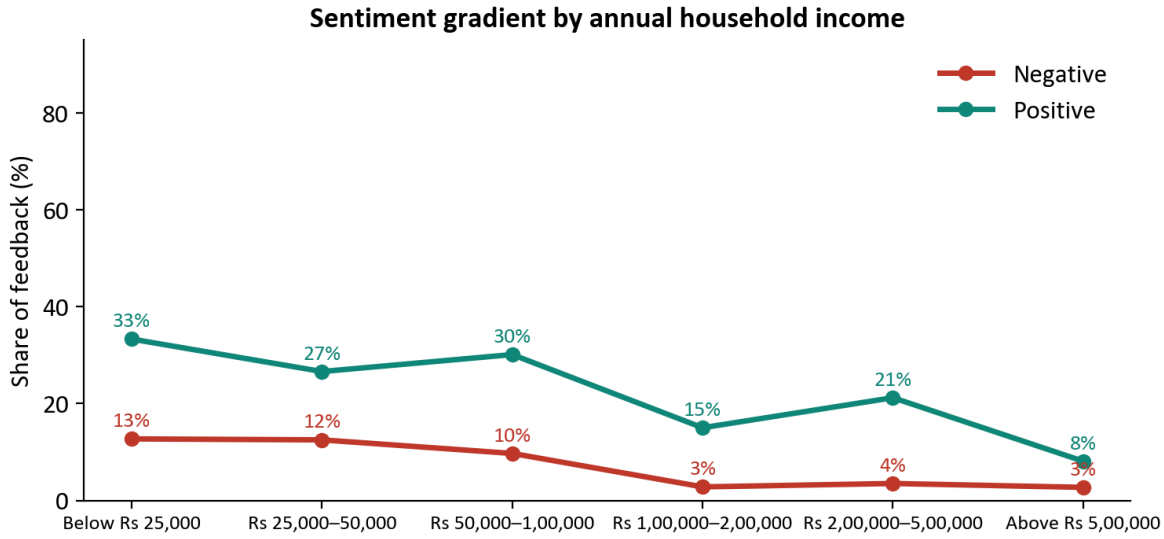
The sharpest finding concerns the relationship between ratings and words. Among 1,180 farmers with both a satisfaction rating and written feedback, 87.4% rated their scheme positively, but only 24.1% wrote about it positively; negative accounts were also more frequent in writing (7.4%) than in ratings

(1.3%). Of the 1,031 farmers who gave a positive rating, 771 (75%) described a mixed, neutral or negative experience, typically acknowledging the benefit while reporting delays, documentation or insufficiency. The McNemar–Bowker test confirms that this asymmetry is systematic ($\chi^2 = 705$, df = 3, n = 1,180, $p < 0.001$). Satisfaction ratings therefore capture gratitude for receiving support far more than the quality of the delivery process.



The gap appears in every major scheme. Crop Insurance has the lowest share of positive written accounts (9.7%), and the Kisan Credit Card the highest (46.9%), although even there positive ratings far exceed positive descriptions. PM-KISAN combines high satisfaction (1.02) with the lowest self-reported impact on farming (0.74), indicating that farmers value its reliability but do not find the fixed instalment large enough to change their farming.

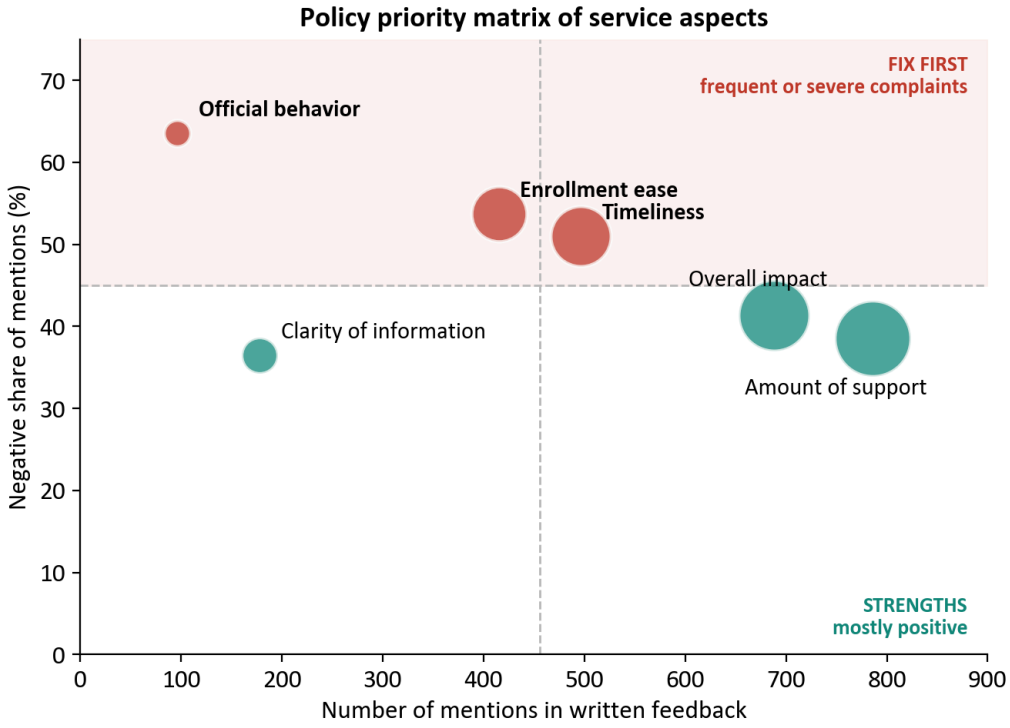
6.4 Equity across farmer groups



Written sentiment differs significantly by education (Cramér's V = 0.20), income (0.17), information source (0.12), scheme (0.13), landholding (0.12) and gender (0.07), all $p < 0.05$, with education and income showing the strongest associations. Lower-income farmers are the most vocal: negative accounts fall from 12.7% among households earning below ₹25,000 to 2.7% among those earning

above ₹5,00,000, while higher-income farmers mostly give neutral, matter-of-fact accounts. The farmers most dependent on scheme support are therefore the ones most likely to report problems with it.

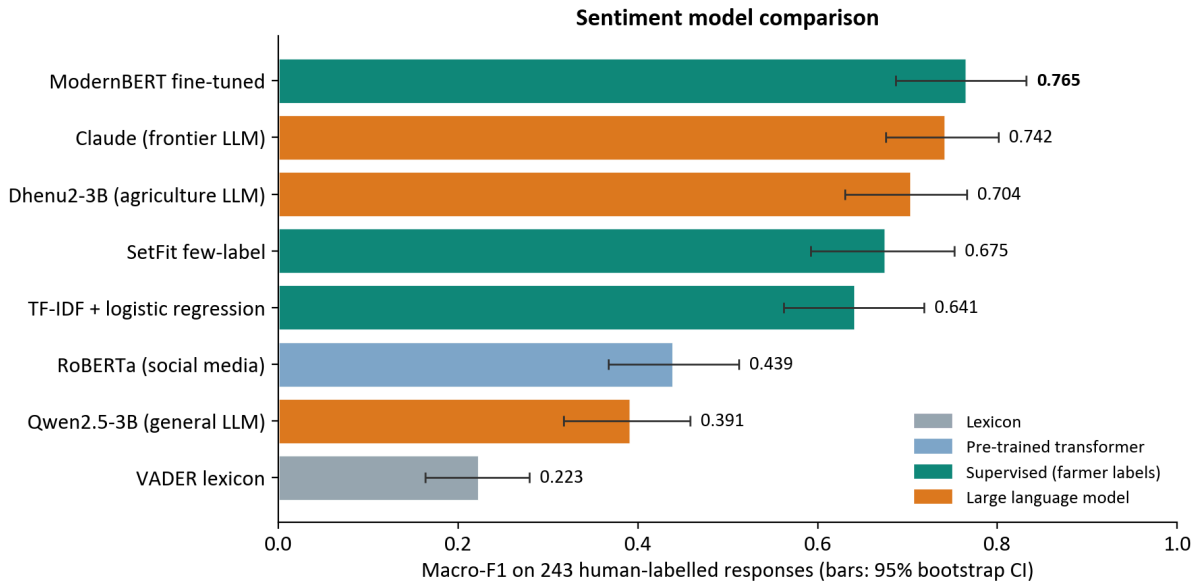
6.5 Policy priority matrix



Plotting each service aspect by how often it is mentioned and how negatively it is described separates priorities clearly. Enrolment ease and timeliness are both frequent and severe and require immediate action; the amount of support and overall impact are mentioned most often and described mostly positively, and are strengths to be preserved.

7. Model Comparison

All approaches were evaluated on the same 243 held-out responses (143 neutral, 76 positive, 24 negative).



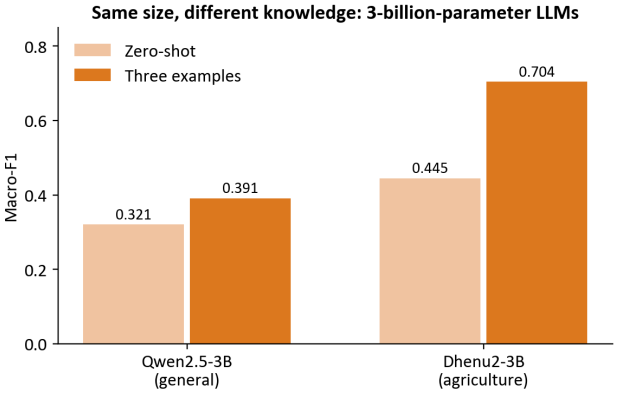
Approach	Accuracy	Macro precision	Macro recall	Macro-F1	95% CI	Cohen's κ
ModernBERT (fine-tuned)	0.835	0.810	0.736	**0.765**	0.687 - 0.832	0.683
Claude (zero-shot)	0.786	0.746	0.787	0.742	0.676 - 0.801	0.620
Dhenu2-3B (3 examples)	0.749	0.682	0.746	0.704	0.63 - 0.766	0.566
SetFit	0.840	0.909	0.647	0.675	0.592 - 0.752	0.675
TF-IDF + logistic regression	0.745	0.708	0.618	0.641	0.562 - 0.718	0.491
RoBERTa (social media)	0.506	0.445	0.471	0.439	0.367 - 0.512	0.103
Qwen2.5-3B (3 examples)	0.457	0.445	0.472	0.391	0.317 - 0.458	0.065
VADER lexicon	0.235	0.215	0.296	0.223	0.163 - 0.279	-0.144

The fine-tuned ModernBERT encoder achieved the highest macro-F1 (0.765) and an accuracy of 83.5%, with a mean of 0.750 ± 0.034 across five seeds. Its advantage over Claude ($p = 0.22$) and SetFit ($p = 1.00$) is not statistically significant, so these three form the leading group; ModernBERT is significantly better than Dhenu2 ($p = 0.017$) and all remaining approaches ($p < 0.01$). A DeBERTa-v3 encoder fine-

tuned under identical settings reached 0.533, showing that the choice of encoder matters on small datasets.

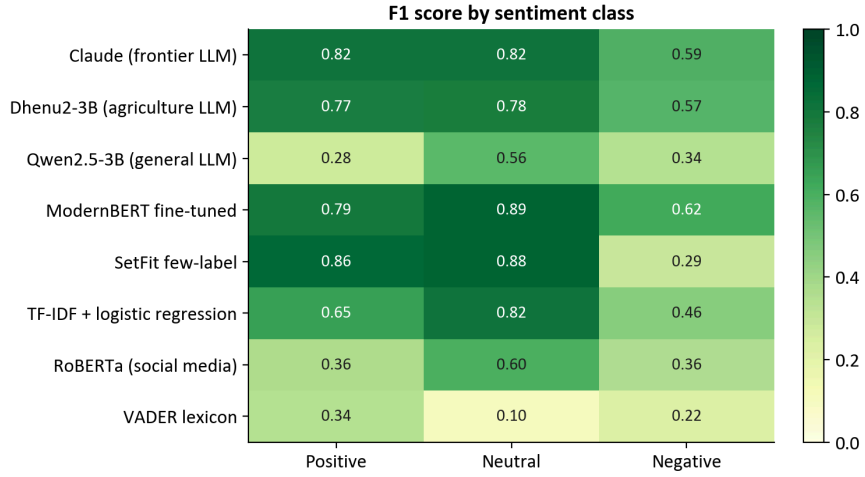
The general-purpose tools perform poorly. VADER agreed with the human labels less often than chance (κ = -0.144) and RoBERTa reached 0.439; the two also agreed only slightly with each other on identical text (κ = 0.155). Sentiment figures produced by these tools on farmer feedback without validation would be misleading.

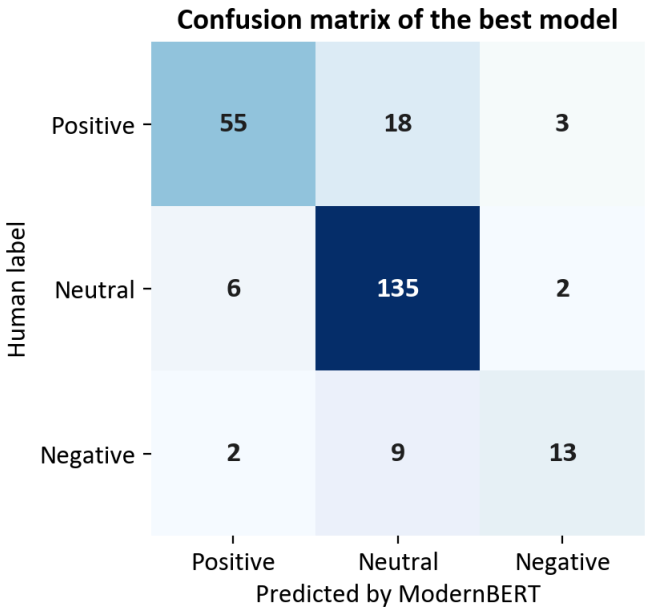
7.1 Domain knowledge in large language models



Dhenu2 and Qwen2.5-3B have the same size, so the difference between them isolates the effect of agricultural training. Without examples Dhenu2 already scored higher (0.445 versus 0.321); with three examples per class the gap widened to 0.313 macro-F1 (0.704 versus 0.391). Domain knowledge also made the agricultural model far more responsive to examples, gaining 0.259 against 0.070 for the general model.

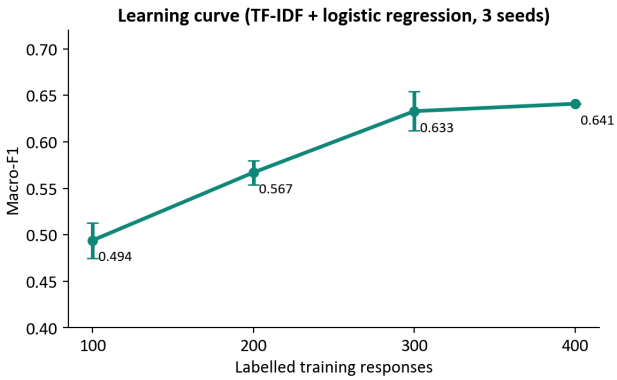
7.2 Performance by sentiment class





Every approach found negative feedback hardest. The best model reached an F1 of 0.62 on negative responses against 0.89 on neutral ones, mainly because 9 of the 24 negative responses were read as neutral. This mirrors the gap reported for negative patient feedback in healthcare [11] and reflects how rarely farmers phrase complaints in strongly negative language.

7.3 Effect of labelled data and aspect-level models



Macro-F1 increased with every additional 100 labelled responses, from 0.494 to 0.641, so further labelling would yield further gains. At the aspect level, a dedicated DeBERTa aspect model detected all 274 human-identified aspect mentions and reached a macro-F1 of 0.433, compared with 201 detected mentions and 0.373 for the keyword method used in the first stage of the pipeline.

8. Discussion

The results show that the largest losses in the scheme pathway occur after awareness. Most farmers know about at least one scheme, but many do not enrol, and those who enrol describe enrolment and payment timing as the weakest parts of their experience. This shifts the policy question from "do farmers know?" to "can farmers get in, and are they paid on time?". The finding complements awareness-centred studies [9] by locating where the barriers act within the same respondents.

The satisfaction gap explains why these problems remain hidden in conventional evaluation. Most farmers give a positive rating, yet most of those same farmers describe mixed experiences in their own words, combining appreciation for the benefit with complaints about delays and procedures. Ratings thus measure whether support arrived, not how well it was delivered, and monitoring that relies on ratings alone will miss the operational failures that matter most to farmers. Importantly, this conclusion holds only when the text is classified by a validated model: the lexicon used in the first stage of the pipeline labelled most responses positive and would have hidden the gap.

Methodologically, the benchmark supports two conclusions from recent literature. First, LLMs can code open-ended survey responses without task-specific training at a level close to strong supervised models [12, 13]: the frontier LLM was statistically tied with the best fine-tuned encoder. Second, small encoders trained on a few hundred in-domain labels remain highly competitive [10, 16] and run cheaply on local hardware, which matters for government use. Lexicon tools and social-media models, still common in agricultural sentiment work [8], are not reliable on farmer feedback.

The comparison between Dhenu2 and Qwen2.5 provides controlled evidence that agricultural domain training improves interpretation of farmer language at a fixed model size. For public agencies that need to run models locally or under data-protection constraints, a small domain-specific LLM given a few labelled examples offers a practical alternative to large general models.

9. Policy Recommendations

The following recommendations are addressed to the Department of Agriculture and Farmers Welfare, Government of Andhra Pradesh, in order of priority.

1. **Review enrolment across all schemes as a single process.** Enrolment ease is among the two most negatively described aspects under every method tested and recurs in the complaint topics; a shared audit of registration, documentation and verification steps at Rythu Bharosa Kendrams would address the common bottleneck rather than treating each scheme separately.
2. **Make disbursement reliability an operational target.** Payment timing accounts for about 44% of complaint text. Publishing payment schedules, sending SMS notifications at each stage and releasing benefits before the sowing season would address the largest single complaint.
3. **Track awareness-to-enrolment conversion as a scheme indicator.** Schemes such as Crop Insurance (9.1%) and PM Krishi Sinchai Yojana (2.2%) need assisted enrolment at the point of application rather than further awareness campaigns.
4. **Commission a dedicated review of Crop Insurance,** which combines the lowest conversion among major schemes, the highest negative feedback and timing-dominated complaints about claim

settlement.

5. **Review the adequacy of the PM-KISAN instalment,** whose problem is low felt impact on farming rather than dissatisfaction with delivery.
6. **Prioritise low-income and marginal farmers** for assisted enrolment and grievance support, since their feedback is consistently the most negative.
7. **Add open-ended questions on enrolment and payment timing to routine monitoring,** and analyse them with models validated on local labelled data, since 87% positive ratings coexist with only 24% positive written accounts.
8. **Do not use unvalidated lexicon tools for policy monitoring.** A small model trained on a few hundred locally labelled responses, or a domain-specific LLM given examples, provides reliable results at low cost.

10. Limitations

- The data cover one district and one survey round; findings are correlational and should be confirmed in other districts and over time.
- Interviews were conducted mainly in Telugu and recorded in English, which may reduce nuance; models operating directly on Telugu text were not evaluated.
- Labels were produced jointly as a single consensus set, so inter-annotator agreement could not be measured; the training labels were AI-suggested and human-verified.
- Negative responses were under-represented in the test set (24 of 243), increasing uncertainty for that class.
- Scheme-level results for small schemes, including Crop Insurance (n = 31), rest on small samples.
- Aspect-level sentiment remains difficult: the best aspect pipeline reached a macro-F1 of 0.38, so aspect findings are reported only where they agree with independent topic evidence.
- Topic models were moderately stable across random seeds, so topic shares are approximate.

11. Conclusion and Future Work

This study shows that farmers' written feedback reveals where welfare-scheme benefits are lost and that satisfaction ratings conceal these losses. Awareness is broadly achieved, but enrolment and timely delivery are the binding constraints, and while 87% of farmers rate their scheme positively, only 24% describe it positively in their own words. The benchmark establishes that such analysis is reliable only with validated models: fine-tuned encoders and capable LLMs perform well, general-purpose lexicon tools do not, and agricultural domain knowledge measurably improves LLM performance.

> **Research publication.** The findings are being prepared as a research paper targeting a Q1 journal, under the guidance of Prof. Sangeetha S. Candidate venues in applied AI for agriculture, rural development and information processing are being shortlisted.

Future work will extend the analysis to additional districts and survey rounds, add independent double annotation and a larger negative class, evaluate multilingual models on the original Telugu responses [24], train a dedicated aspect-level model on a larger labelled set, and develop an interactive dashboard for the Department of Agriculture.

Project links

Source code and results: https://github.com/singhkrish2006/farmer-scheme-policy-intelligence
Final project presentation:
https://drive.google.com/drive/folders/1Ea17IGBCi2gmo6G_VSptUSgVGAnb_yEK?usp=drive_link

The raw survey and labelling worksheets are not published because they contain personal information of the participants.

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Appendix

A. LLM instructions

> You label farmers' feedback about Indian government welfare schemes. Reply with exactly one word: Positive, Negative or Neutral.
> Positive = mainly expresses satisfaction, benefit, relief or approval.
> Negative = mainly expresses a problem, delay, frustration or complaint.
> Neutral = factual with no clear tone, or genuinely mixed with no dominant tone.
>
> Few-shot: 3 training examples per class, seed 42; greedy decoding

B. Image credits

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